I. STOP-AND-WAIT ARQ EFFICIENCY PROOF PROCEDURE

General process is

- **Transmits one data frame.**
- **Waits for an acknowledgment (ACK) from the receiver.**
- **Transmits the next frame only after receiving the ACK.**
- **Retransmits the frame if an ACK is not received before the timeout.**

Symbol	Definition	Formula
L	Data Frame Length	bits
R	Transmission Rate	bits/s
T_t	Frame Transmission Time	$\frac{L}{R}$
T_p	One-Way Propagation Delay	$\frac{d}{v}$
T_{ACK}	ACK Transmission Time	seconds
a	Propagation-to-Transmission Ratio	$\frac{T_p}{T_t}$
η	Normalized Throughput	Useful time / Total time
d	The Distance Between Sender and Receiver	
v	The Distance Between Sender and Receiver	

Channel utilization is the fraction of time the sender spends transmitting useful data, therefore we have

$$\eta = \frac{Useful\ Transmission\ Time}{Total\ Cycle\ Time} \rightarrow \eta = \frac{T_t}{T_c}$$

We reach efficiency for one successful frame, then we go for complete cycle time as follows, neglecting the processing delays, we can formulate the process like below

$$T_{cycle} = T_t + T_p + T_{ACK} + T_p \ \rightarrow \ T_{cycle} = T_t + 2T_p + T_{ACK}$$

Then

$$\eta = \frac{1}{1 + 2\frac{T_p}{T_t} + \frac{T_{ACK}}{T_t}} \rightarrow assuming\ T_{ACK} = 0 \rightarrow \eta = \frac{1}{1 + 2\frac{T_p}{T_t}} \rightarrow we\ know\ a = \frac{T_p}{T_t}$$

Therefore

$$\boxed{\eta = \frac{1}{1 + 2a}}$$

II. GO-BACK-N ARQ EFFICIENCY PROOF PROCEDURE

As we pursued the above procedure we directly and briefly start the math proof, as depicted below

$$\eta = \frac{Useful\ Transmission\ Time}{Expected\ Total\ Time} = \frac{T_t}{E[T]}$$

$$[K_f] = \frac{P}{1-P}$$

$$\begin{aligned} E[T] &= T_t + E[K_f]NT_t \\ &= T_t + E[K_f]NT_t \\ &= T_t\left(1 + \frac{NP}{1-P}\right) \\ &= T_t\frac{1-P+NP}{1-P} \\ &= T_t\frac{1+(N-1)P}{1-P} \end{aligned}$$

$$\eta = \frac{T_t}{T_t\frac{1+(N-1)P}{1-P}}$$

$$\boxed{\eta = \frac{1-P}{1+(N-1)P}}$$

III. SELECTIVE REPEAT ARQ EFFICIENCY PROOF PROCEDURE

As we pursued the above procedure we directly and briefly start the math proof, as depicted below

$$\eta = \frac{\text{Useful Transmission Time}}{\text{Expected Total Transmission Time}} = \frac{T_t}{E[T]}$$

$$\Pr(K = k) = P^{k-1}(1 - P)$$

$$E[K] = \sum_{k=1}^{\infty} kP^{k-1}(1 - P)$$

$$= (1 - P)\sum_{k=1}^{\infty} kP^{k-1}$$

$$= (1 - P)\frac{1}{(1 - P)^2}$$

$$= \frac{1}{1 - P}$$

$$E[T] = E[K]T_t$$

$$= \frac{T_t}{1 - P}$$

$$\eta = \frac{T_t}{T_t/(1 - P)}$$

$$\boxed{\eta = 1 - P}$$